

Week 8

ITIL Process

COS40006 Computing Technology Project B

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COS40006 Computing Technology Project B

- 01** Scrum Revisited
- 02** Scrum Master
- 03** Code of Conduct
- 04** Employability
- 05** Resume & Interview Prep
- 06** Transition to Employment

- 07** CapEx
- 08** ITIL Process
- 09** Ethical AI
- 10** Project Closure & Handover
- 11** HDR Pathway
- 12** Workplace Safety

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Bonus:
CapEx



Capital Expenditure

['ka-pə-təl ik-'spen-di-chərz]

Funds used by a company to acquire, upgrade, and maintain physical assets such as property, plants, buildings, technology, or equipment.

Understanding Capital Expenditure (CapEx) for Tech Careers

Why this is important to you:

- Impacts budgets for tech tools, infrastructure, and R&D.
- Affects project feasibility and timelines.
- Helps in understanding how companies allocate resources for innovation.

Understanding Capital Expenditure (CapEx) for Tech Careers

CapEx

= funds used by a company to acquire, upgrade & maintain physical assets (e.g., equipment, buildings, technology).

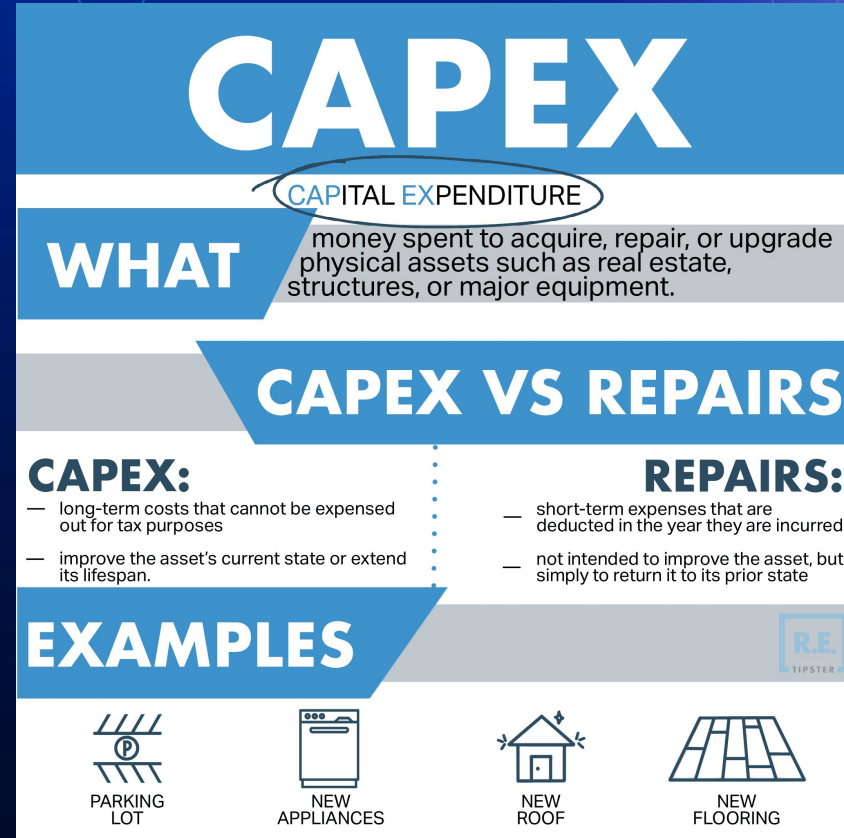
Types of CapEx

Growth CapEx – Expanding capacity or capabilities

- e.g., Adding a new data center, expanding network infrastructure.

Maintenance CapEx – Replacing/repairing existing assets

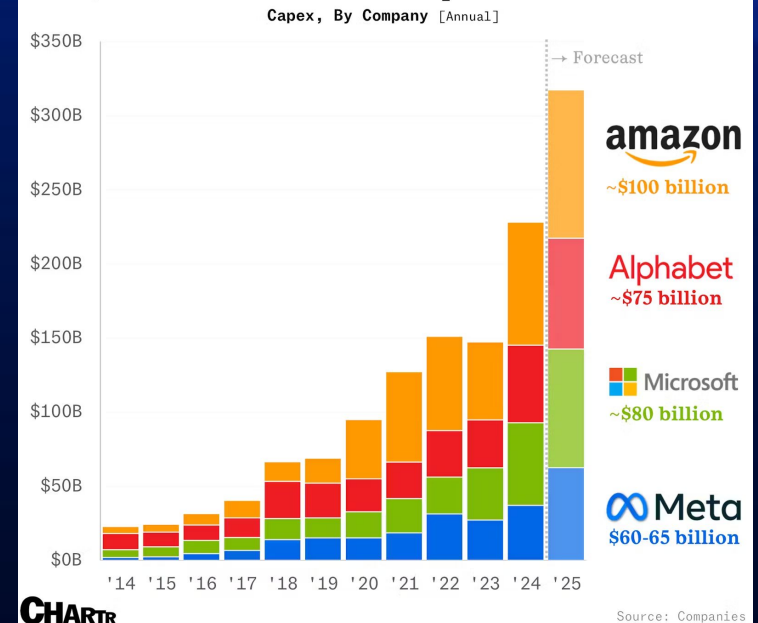
- e.g., Replacing old servers, upgrading outdated lab equipment.



CapEx in Tech Industry

- Hardware Purchases: Servers, workstations, IoT devices.
- Software Licenses (perpetual): Paid upfront and owned permanently.
- Infrastructure: Networking hardware, cooling systems in data centers.
- Lab & R&D Equipment: Robotics kits, 3D printers for prototyping.

4 Big Tech Companies Plan On Spending \$315+ Billion On Capex This Year



CapEx Approval Process

- Proposal: Justify the need (ROI, efficiency gains).
- Budgeting: Fits within annual capital budget.
- Approval: Senior management/board sign-off.
- Procurement: Vendor selection, purchase.
- Deployment: Install & integrate into operations.



Impact on Projects

- Positive: Enables new capabilities, faster delivery.
- Negative: Budget delays, longer procurement cycles.

IT Example:

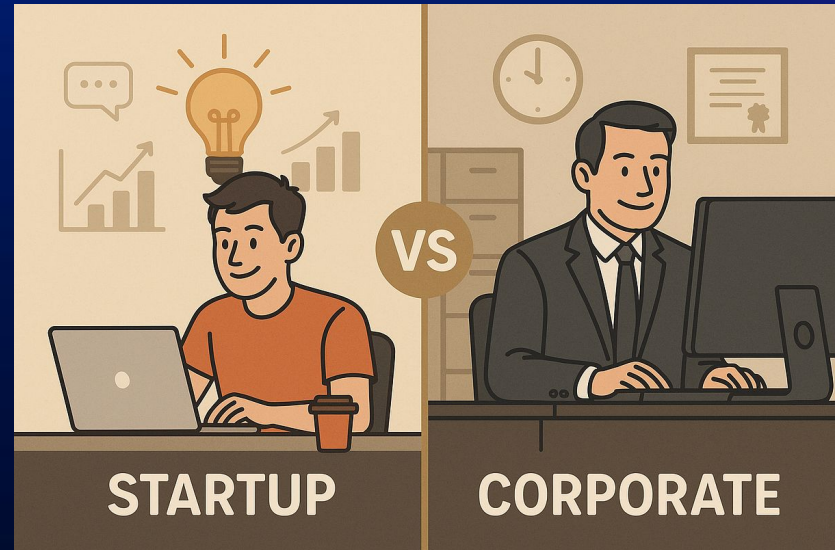
Waiting 3 months for GPU purchases can delay
AI model training.



CapEx in Startups vs Corporates

- **Startups** : Leaner budgets, prefer OpEx (cloud, SaaS) to avoid high upfront costs.
- **Corporates** : Can afford large CapEx for long-term savings.

Case: Netflix's shift from in-house servers (CapEx) to AWS cloud (OpEx).



Practical Skills for Graduates

- Understand basic ROI calculations for CapEx requests.
- Be familiar with TCO (Total Cost of Ownership).
- Learn how procurement cycles work in tech companies.
- Know when to recommend OpEx over CapEx for flexibility.

— CapEx —



Physical assets
like data centers



Hardware



Licensing
for software



Protections like
patents, trademarks
and copyrights



Bulk of development
work upfront

— OpEx —



Subscriptions and
cloud based expenses



Integration Platform
as a Service



Software
as a Service
solutions



Post implementation
costs and
maintenance

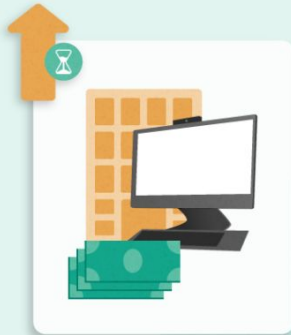
Capex

vs.

Opex

Large expenditures that a company expects to influence production and profitability.

Daily, weekly, monthly and annual expenses that keep the organization running.



CapEx Examples



IT Infrastructure
(servers, routers,
switches, software,
network cabling, etc.)



**IT Equipment for
Employees**



**Real Estate
Purchase**



**Real Estate
Renovations**



**Asset Upgrade
Costs**



**Infrastructure
Repairs**

OpEx Examples



**Business-related
Operating Costs**
(rent, utilities, salaries,
legal fees, etc.)



**Datacenter or
Off-premises
Cloud Costs**



**Software,
Application,
or Service
Subscription Fees**



**Webhosting
and Domain
Licenses**



**Software and
Service Support**



**Research and
Development
Expenses**

Tip: In tech, cloud vs. on-premises decisions often hinge on CapEx vs OpEx.



CapEx

Capital Expenses

VS



OpEx

Operating Expenses

**Buying your own
car**

**Hiring a Grab
driver**

Real Business Examples

QUIZ.com

What is

ITIL

What is

ITIL

Information Technology

What is

ITIL

Infrastructure

What is

ITIL

Library

What is ITIL?

ITIL = Information Technology Infrastructure Library

A **framework** of best practices for managing IT services.

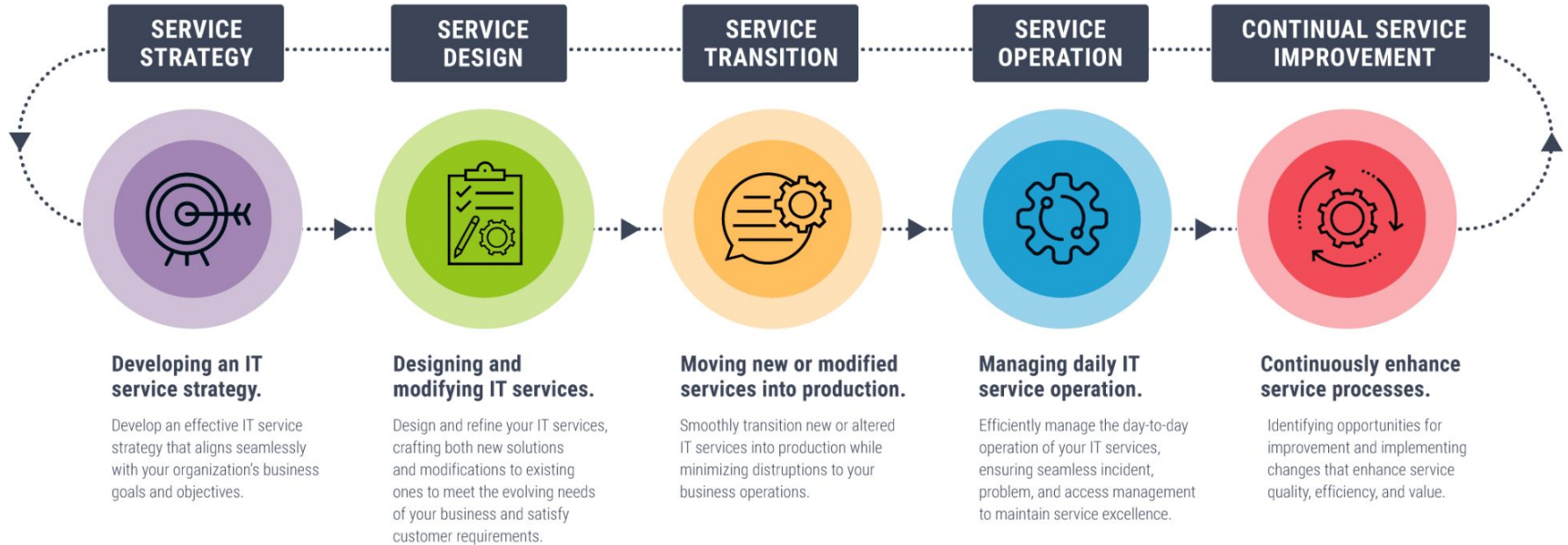
It focuses on **aligning IT with business goals** and improving the efficiency and effectiveness of IT service delivery.

What is ITIL?

ITIL framework organizes IT service management into 5 core stages:

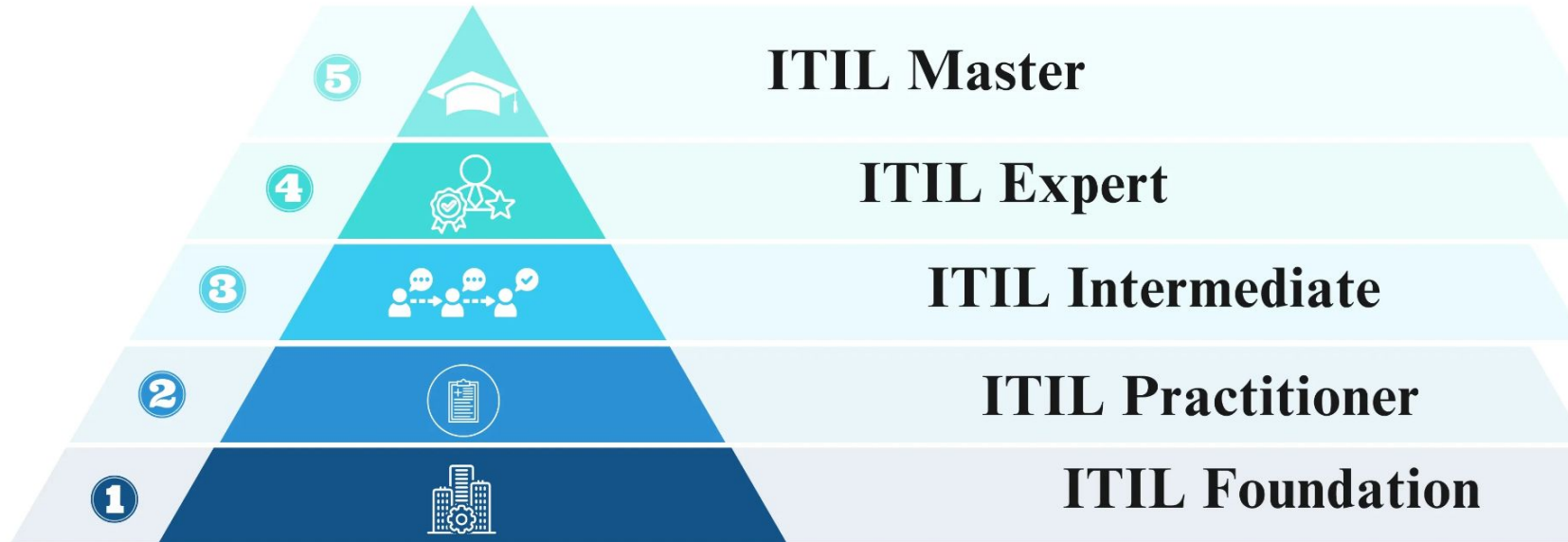
1. Service Strategy
2. Service Design
3. Service Transition
4. Service Operation
5. Continual Service Improvement

ITIL Processes



Sources: [axelos.com](https://www.axelos.com) and [vivantio.com](https://www.vivantio.com)

ITIL Certification Levels



ITIL Master*

ITIL Practice Manager (PM)**

Monitor, Support and Fulfil

SD INM PRM SRM MEM

Plan, Implement and Control

CE RM SCM DM ITAM

Collaborate, Assure and Improve

CI SLM RSM ISM SM

ITIL Managing Professional (MP)

Direct, Plan
and Improve

High-velocity IT

Drive Stakeholder
Value

Create, Deliver
and Support

ITIL Strategic Leader (SL)

Digital and IT
Strategy

ITIL Extension modules

Acquiring and
Managing
Cloud Services

Business
Relationship
Management

Sustainability
in Digital
and IT

IT Asset
Management

ITIL Foundation

* ITIL4 Master will be awarded to candidates who achieve the Practice Manager (PM), Managing Professional (MP), and Strategic Leader (SL) designations.

** To be awarded the Practice Manager designation, a candidate must achieve ITIL MP CDS certificate and ANY five practice-based certifications, OR ITIL MP CDS certificate and any ONE of the pre-bundled courses. These may include the ITAM and BRM extension modules or any practices from the Practice Manager track.

Conclusion

The background is a deep blue gradient. It features several abstract elements: a large, dark, wavy shape on the left side; a series of light blue, glowing hexagonal outlines in the top right corner; and a series of light blue, glowing hexagonal outlines in the bottom left corner. A prominent, glowing, wavy line composed of many small, light blue dots runs diagonally across the bottom right corner.

Key Takeaways

ITIL Core Practices You'll Encounter in Jobs

- Incident Management: Quick restore of services.
- Problem Management: Find root causes.
- Change Enablement: Structured, low-risk changes.
- Service Desk: Single point of contact for users.
- Knowledge Management: Documentation + reuse of solutions.



Kaodim!

Any questions?

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Additional resources

- <https://www.atlassian.com/itsm/itil>
- <https://www.youtube.com/watch?v=Zt6GNAIoUsY>
- <https://youtu.be/fpSYNahuWWM?si=WbTu2ZL4uqpK0Fdp>